

behaviour of edge systems under failure conditions. They noticed that application data reporting errors occurred when network packet loss reached 10% of total packets. The implementation of automated recovery scripts reduced system errors by nearly 30%. In another example, engineers at an industrial site performed tests that duplicated sensor breakdowns on their small factory production area. The edge computing modules operated continuously because they maintained system operation while critical data streams recovered within five seconds.

The testing of IoT devices needs certain methods to handle distinct challenges. Inadequate testing procedures often lead to permanent damage to devices and create safety hazards. The placement of multiple devices in inaccessible locations complicates physical maintenance operations. An uncontrolled chaos test in a smart home system could lead to a failed smart lock that prevents homeowners from accessing their property. The restart of glucose monitors during measurement operations can create dangerous situations for patients who use these devices. Industrial networks can shut down completely when they encounter any level of system disruption.

The implementation of chaos engineering for IoT systems requires both strategic planning and innovative solutions. Engineers should perform system vulnerability tests, which ensure operational safety and reliability for real world deployment. The risk assessment process needs tested and accurate methods to protect both system devices and their users from harm.

Lack of open source tools for chaos engineering in IoT systems

Over time, many open source tools have been created to make chaos engineering easier in cloud systems. These tools work well for cloud-based

Table 2: Chaos engineering overview

Aspect	Description
Definition	Creating artificial system breakdowns
Origin	Stems from Netflix Chaos Monkey
Implementation	Microsoft, Google, Amazon, etc
Edge	Physical devices, remote, critical tasks
Tools	µChaos, ZephyrOS, fault injection, controlled tests
Benefits	Find vulnerabilities, improve resilience
Challenges	Risk to devices, complex reboots, slower recovery

servers and containers but do not work as effectively on Internet of Things (IoT) devices, such as sensors, smart cameras and small robots. This is because most IoT devices do not use Kubernetes or Docker systems. Their operating systems are small or run directly on chip-level firmware rather than standard systems.

Most IoT devices are powered by microcontrollers with only a few megabytes of memory. This memory is much smaller than even what basic smartphones possess, which makes it hard to run chaos experiments as they require a lot of space to simulate failures.

However, IoT system testing methods are evolving through research

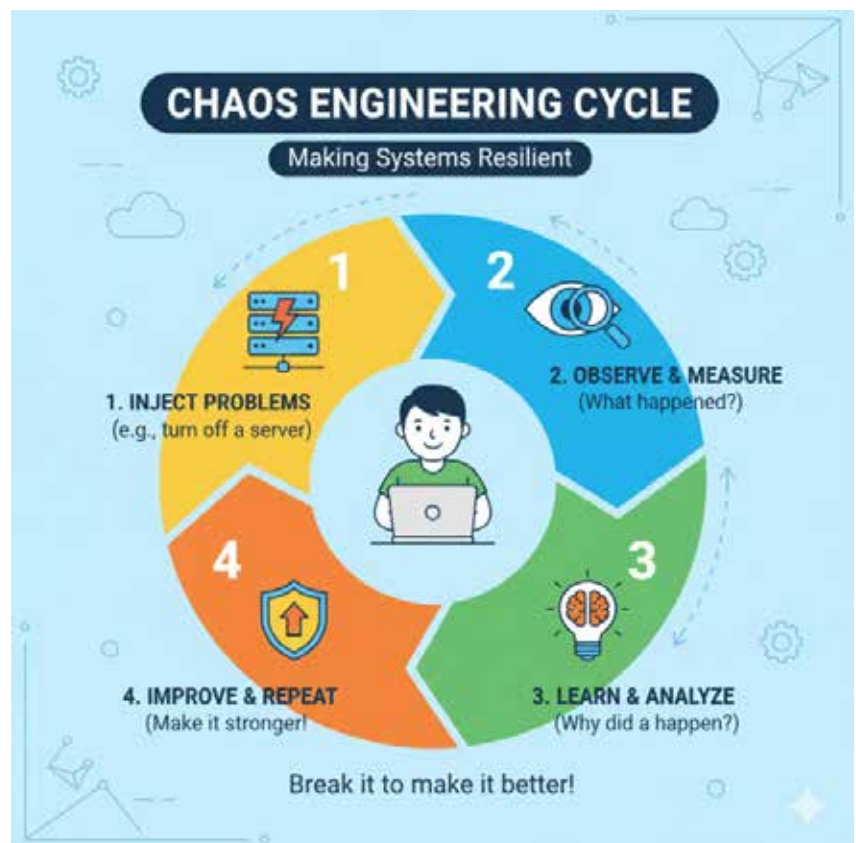


Figure 1: Chaos engineering cycle